

— STEP TODAY, POWER TOMORROW —

FOOTSTEP POWER GENERATION

Harnessing Energy from Human Footsteps
for a Sustainable Future



HARNESS
MOTION



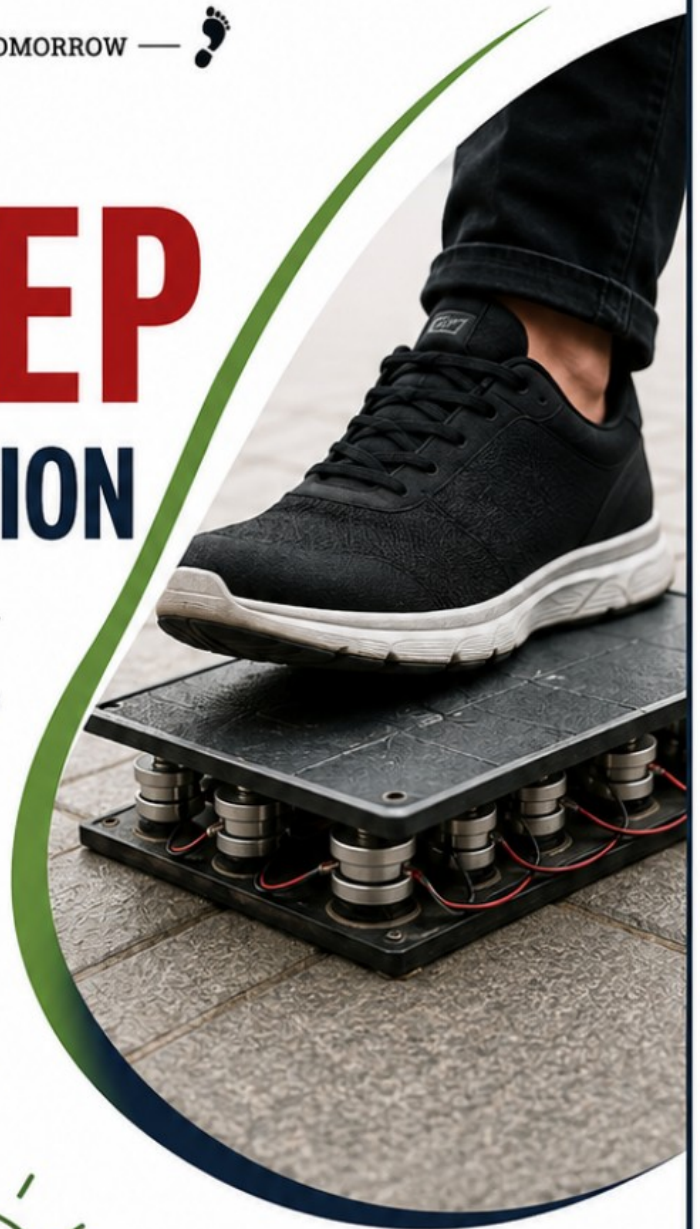
GENERATE
POWER



STORE &
UTILIZE



CLEAN &
SUSTAINABLE



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ABSTRACT

The main aim of this project is to develop a using Arduino Uno. The main aim of this project is to develop much cleaner cost effective way of power generation method, which in turns helps to bring down the global warming as well as reduce the power shortages and dependency on non-renewable.

In this foot step power generation project is to convert foot step, walking and running energy into electrical energy. It is used to generate electricity from by walking on foot step. The need of electrical energy is increasing day by day. But power generation conventional resources not enough for a total demand of electrical energy.

Many researchers are working on non-conventional ways of electrical power generation. Foot step power generation system is also a non-conventional electrical energy production system. It coverts mechanical energy of foot steps into electrical energy by using transducers.

CONTENTS

ACKNOWLEDGEMENT	i
ABSTRACT	ii
CONTENTS	iii-iv
LIST OF FIGURES	v
CHAPTER 1	
1. INTRODUCTION	1
1.1 Overview	1
1.2 Problem Statement	2
1.3 Objectives	3
CHAPTER 2	
2. LITERATURE SURVEY	4-5
CHAPTER 3	
3. HARDWARE AND SOFTWARE REQUIREMENTS	6
3.1 Hardware Requirements	6-11
3.2 Software Requirements	11-12
CHAPTER 4	
4. SYSTEM DESIGN	13
4.1 Block Diagram	13
4.2 Working and Principle	14

CHAPTER 5

5. RESULTS AND DISCUSSION 15

5.1 Result 15

CHAPTER 6

6. ADVANTAGES AND DISADVANTAGES 16

6.1 Advantages 16

6.2 Applications 16

6.3 Disadvantages 16

CHAPTER 7

7. CONCLUSION

7.1 Conclusion 17

CHAPTER 8

8. REFERENCE 18

LIST OF FIGURES

FIGURE NO	FIGURE NAME	PAGE NO
3.1	Cell holder	9
3.2	BC547 Transistor	11
3.3	1N4007 Diode	11
3.4	10uF Capacitor	12
3.5	Jumper wire	13
3.6	Hook up wire	14
3.7	Mini Bread board	14
3.8	LCD Display	15
3.9	Arduino Uno	16
3.10	35mm piezo ceramic element	17
3.11	Transparent Acrylic sheet	18
3.12	Arduino IDE	20
4.1	Block diagram	20
4.2	Schematic Representation	21

4.3	Circuit Connections	22
5.1	Project Prototype	23

CHAPTER 1

INTRODUCTION

1.1 Overview

The SUN is the most important source of energy for the life on the earth either in direct or derivative form. Dependency on non-renewable sources decreasing these sources day by day and in near future it may get exhausted completely. Hence it is required to explore for alternative sources and shift our dependency on renewable sources. This will conserve non-renewable sources and produce clean energy. These renewable sources include solar cells (Solar energy), wind mills (Wind energy), geothermal power plants (Geothermal energy), tidal turbine (Tidal energy) etc. Solar power provides a considerable amount of energy per area and volume, but unfortunately is limited to applications that are actually sunlit. In recent years, piezoelectric technology has been used to convert the available mechanical energy potential in pavement into electrical energy, which has been widely used in roadways and bridges. The voltage produced from piezoelectric material varies with time and results in an alternate current (AC) signal, which causes the direct and inverse piezoelectric effect, respectively. Piezoelectric technology has advantages over the other energy harvesting technologies in infrastructure: it can be used for both energy harvesting and pavement structure sensing, which will be quite beneficial for smart infrastructure. The challenge of using piezoelectric technology for energy harvesting is to integrate the piezoelectric transducer with the structure of roadway and bridges. The energy harvesting performance of piezoelectric transducers (PZTs) is affected by material, geometry design of transducer, and external loading.

We utilize a large part of our muscular energy for moving from one place to other and also the infrastructure like roads, railways, runway bears a large amount of mechanical strain energy. This energy i.e. muscular or mechanical strain on various infrastructures gets wasted. But it is possible to convert this mechanical energy in to electrical pulse form with the help of piezoelectric transducers. These electrical pulses, which are alternating in nature, can be directly utilized or may be captured by a storage device for further utilization. Efforts have been put in this work to harvest energy from mechanical stress using the principle of piezoelectric energy conversion. For a harvesting system of constant thickness, the generated power increases with increase in applied force. The output power of harvester depends on increase in the thickness. The output power obtained from piezoelectric generators depends

on various factors like which piezoelectric sensor has been used, its packing density, type of strain applied to it, electronic circuitry to process the pulse generated, storage device, and load connected to it. When a simple rectifier is used the output power generated greatly depends upon the load connected. The important criteria for maximizing the output power are to match the optimal load of the harvester to that of converter circuit. Several techniques are available for converting mechanical vibration energy to electrical energy. The most prevalent methods among them are electrostatic, electromagnetic and piezoelectric conversion. A majority of current research has been done on piezoelectric conversion due to low complexity of its analysis and fabrication. Most of research however has targeted a specific device scale. The latest advancement in the micro-electromechanical systems (MEMS) and wireless technology, the portable electronics and wireless sensors are in great demand.

Previous studies have investigated the feasibility of applying piezoelectric energy harvester on roadways and bridges to generate electric energy effectively for different uses. However, few works have been conducted to investigate the potential of using piezoelectric energy harvesting at airport for green energy solutions. Airfield pavements are designed to carry repeated loading of heavy aircraft. Therefore, research is needed to explore the potential of harvesting energy in airport infrastructure using piezoelectric technology.

1.2 Problem Statement

Proposal for the utilization of waste energy of foot power with human locomotion is very much relevant and important for highly populated countries like India and China where the roads, railway stations, bus stands, temples, etc. are all over crowded and millions of people move around the clock. This whole human/bio-energy being wasted if can be made possible for utilization it will be great invention and crowd energy farms will be very useful energy sources in crowded countries. Walking across a "Crowd Farm," floor, then, will be a fun for idle people who can improve their health by exercising in such farms with earning. The electrical energy generated at such farms will be useful for nearby applications. The creation of new source of perennial environmentally acceptable, low cost electrical energy as a replacement for energy from rapidly depleting resources of fossil fuels is the fundamental need for the survival of mankind. We have only about 25 years of oil reserves and 75 – 100 years of coal reserves. Resort to measure beginning of coal in thermal electric stations to serve the population would result in global elementic change in leading to worldwide drought and decertification. The buzzards of nuclear electric-stations are only too will. Now electric

power beamed directly by micro-wave for orbiting satellite. Solar power stations (S.P.S) provide a cost-effective solution even though work on solar photo voltaic and solar thermo electric energy sources has been extensively pursued by many countries. Earth based solar stations suffer certain basic limitations.

1.3 OBJECTIVES

- The main aim of this project is to develop much cleaner cost effective way of power generation method, which in turns helps to bring down the global warming as well as reduce the power shortages and dependency on non-renewable.
- In this foot step power generation project is to convert foot step, walking and running energy into electrical energy. It is used to generate electricity from by walking on foot step.
- The need of electrical energy is increasing day by day. But power generation conventional resources not enough for a total demand of electrical energy.
- Many researchers are working on non-conventional ways of electrical power generation. Foot step power generation system is also a non-conventional electrical energy production system.
- It coverts mechanical energy of foot steps into electrical energy by using transducers.

CHAPTER 2

LITERATURE SURVEY

Ref.1 Footstep Power Generation Using Piezo Electric Transducers.

Piezoelectric ceramics belong to the group of ferroelectric materials. Ferroelectric materials are crystals which are polar without an electric field being applied. The piezoelectric effect is common in piezo ceramics like PbTiO₃, PbZrO₃, PVDF and PZT. The main component of the project is the piezoelectric material. The proper choice of the piezo material is of prime importance. For this, an analysis on the 2 most commonly available piezoelectric material - PZT and PVDF, to determine the most suitable material was done.

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Ref.2

Generated electricity stored in Batteries.

Human activity is over loading our atmosphere with Carbon dioxide and other global warming emissions which causes harmful effects. Energy demand also increasing now-a-days and there is a need for inventing alternative source of energy to cater the needs of human requirements. This paper explores a positive impact that is generating electricity from the waste energy. When a person walked on road surface, there is a loss of energy in the form of impact, vibration etc. transfer from his weight.

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CHAPTER 3

HARDWARE AND SOFTWARE REQUIREMENTS

3.1 Hardware Requirements

1. Cell holder
2. BC547 transistor (PNP)
3. 1N4007 diode
4. 10uF Capacitor
5. Jumper wire
6. Hook up wire
7. Mimi bread board
8. LCD display
9. Arduino Uno
10. 35mm piezo ceramic element
11. Transparent Acrylic sheet.

3.2 Software Requirement

Arduino IDE

Hardware Requirements

Cell Holder :



Fig 3.1: Cell Holder

An 18650 battery holder securely houses one or more 18650 lithium-ion batteries. These holders ensure a reliable electrical connection and provide physical protection to the batteries, making them ideal for various applications ranging from flashlights to power banks and DIY electronics projects.

BC547 transistor (PNP)

The BC547 is a bipolar junction transistor (BJT) with a Negative-Positive-Negative (NPN) bias configuration. It has three terminals: the emitter, collector, and base. The BC54

transistor is often used for switching, amplification, and signal processing in low- and medium-power electronic applications.



Fig 3.2 : BC547 Transistor

The BC547 has two operational statuses: forward bias and reverse bias. In forward bias, current can pass when the collector and emitter are connected. In reverse bias, the transistor acts as a disconnect switch and current cannot pass.

1N4007 Diode



Fig 3.3 : 1N4007 Diode

A 1N4007 diode is a silicon rectifier diode that converts alternating current (AC) to direct current (DC) in electronic circuits. A 1N4007 diode is a PN-junction rectifier that allows current to flow in one direction. It's used in AC-to-DC rectifier circuits and bridge rectifier circuits.

The diode's unidirectional conductivity allows it to convert AC into pulsed DC. In forward-biased conditions, the diode allows current to flow in, and in reverse-biased conditions, it blocks current. This protects circuits from reverse opposition and helps convert AC to DC.

10uF Capacitor

A 10uF capacitor is a device that stores electrical energy in an electric field. The "uF" stands for microfarad, which is a unit of capacitance. One microfarad is equal to 10^{-6} farads.

Here are some more details about capacitors: Capacitors store electrical energy by accumulating electric charges on two closely spaced surfaces that are insulated from each other. Capacitors are used in many applications, including power supplies, audio amplifiers, home theater systems, and hobby projects.



Fig 3.4 : 10uF Capacitor

Jumper wire

Jumper wires are electrical wires with connector pins on each end that are used to connect two points without soldering. They are often used with breadboards and other prototyping tools to make it easy to change a circuit. Here are some things to know about jumper wires: There are three types of jumper wires: female-to-female, male-to-male, and male-to-female. Male-to-female jumper wires are the most common.

Jumper wires come in different colours, but the colours don't indicate a different function. However, colours can be used to differentiate between types of connections, such as ground or power.



Fig 3.5 : Jumper wire

Hook up wire

Hook-up wire is a type of insulated copper wire that's used for low-voltage, low-current applications. It's often used in enclosed spaces, like electronic equipment, and is also

used in automotive, marine, and building construction applications. Here are some things to know about hook-up wire: Hook-up wire can be solid or stranded, and the copper can be bare or tinned. Tinned copper means the wire has a tin coating.



Fig 3.6 : Hook up wires

Mini bread board

A breadboard is a tool used for building and testing electronic circuits without soldering. It has a grid of holes where components can be inserted and connected. Breadboards are commonly used for prototyping and experimenting with circuits.

No Soldering Required: Components are inserted into the board and connected by metal clips, eliminating the need for soldering. • Reusable: Breadboards can be used multiple times for different projects. • Grid of Holes: Offers a grid of holes that are internally connected in rows and columns to facilitate easy connections between components. • Power Rails: Typically includes two long rows (often marked with red and blue lines) for distributing power (positive and negative) across the board.

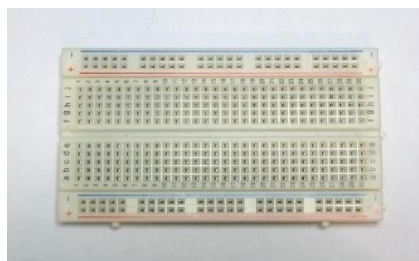


Fig 3.7 : Mini bread Board

LCD Display

An LCD screen is an electronic display module that uses liquid crystal to produce a visible image. The 16×2 LCD display is a very basic module commonly used in DIYs and

circuits. The 16×2 translates a display of 16 characters per line in 2 such lines. In this LCD, each character is displayed in a 5×7 pixel matrix. Liquid crystals do not emit light directly but instead use a backlight or reflector to produce images in colour or monochrome. Layers of a reflective twisted liquid crystal display: Polarizing filter film with a vertical axis to polarize light as it enters.



Fig 3.8 : LCD display

Arduino Uno

The Arduino Uno is a microcontroller board based on the ATmega328P. It is one of the most popular and beginner-friendly development boards for electronics and programming projects. The Arduino Uno is widely used for prototyping and educational purposes, enabling users to create interactive and automated systems.

Arduino Uno Features:

1. Easy to use and program
2. Supports multiple programming languages
3. Open-source platform
4. Affordable
5. Portable



Fig 3.9 : Arduino Uno

35mm piezo ceramic element

A piezoelectric ceramic is a smart material that converts a mechanical effect (such as pressure, movement, or vibration) into an electrical signal and vice versa.

This 35mm piezo plate element are used various electronic Project like foot generated energy. 35mm wide brass piezoelectric discs. These are perfect for using as touch-sensors, electronic drum triggers, buzzers, or making your own contact mics. To use, simply solder positive to the centre and ground to the brass disc.



Fig 3.10 : 35mm piezo ceramic element

Transparent Acrylic sheet

A transparent acrylic sheet is a flat, rigid piece of transparent or coloured thermoplastic plastic that is often used as an alternative to glass. Acrylic sheets are also known as plexiglass or acrylic glass. They have many advantages over glass.

Acrylic sheets are much stronger than glass and have high impact resistance. Acrylic sheets are about half the weight of glass. Acrylic sheets have better weathering properties than many other transparent plastics. Acrylic sheets are easy to work with, fabricate, and bond with adhesives and solvents.



Fig 3.11 : Transparent Acrylic sheet

Software Requirement

ARDUIUNO IDE

The Arduino IDE (Integrated Development Environment) is a software application used for writing, compiling, and uploading code to Arduino boards. It supports programming

in C/C++ and provides an easy-to-use interface with features like code highlighting, debugging tools, and library management.

The IDE allows users to interact with various Arduino boards, such as Arduino Uno, Arduino Nano, and Arduino Mega, and is widely used for prototyping and experimenting with electronic projects. It is available for Windows, macOS, and Linux operating systems.

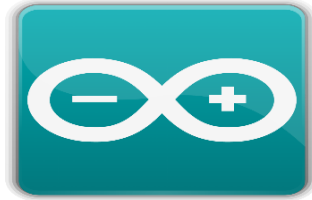


Fig 3.12 : Arduino IDE

CHAPTER 4

SYSTEM DESIGN

4.1 Block Diagram

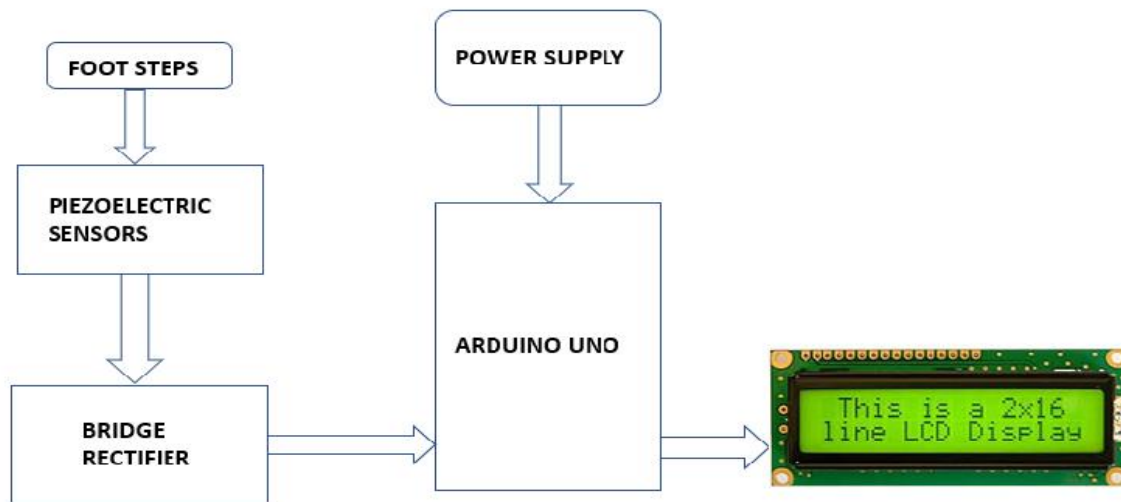


Fig 4.1: Block Diagram

4.2 Working and Principle

Principle

- Footstep power generation systems harness the kinetic energy generated by human footsteps and convert it into electrical energy.
- Piezoelectric materials generate electricity when subjected to mechanical stress, such as pressure or vibration.
- Piezoelectric materials generate electricity when subjected to mechanical stress, such as pressure or vibration.
- The generated electrical energy is typically stored in a battery or capacitor for later use.

Working

The piezoelectric material converts the pressure applied to it into electrical energy. The source of pressure can be either from the weight of the moving vehicles or from the weight of the people walking over it. The output of the piezoelectric material is not a steady one. So a bridge circuit is used to convert this variable voltage into a linear one. Again an AC ripple filter is used to filter out any further fluctuations in the output. The output dc voltage is

then stored in a rechargeable battery. As the power output from a single piezo-film was extremely low, combination of few Piezo films was

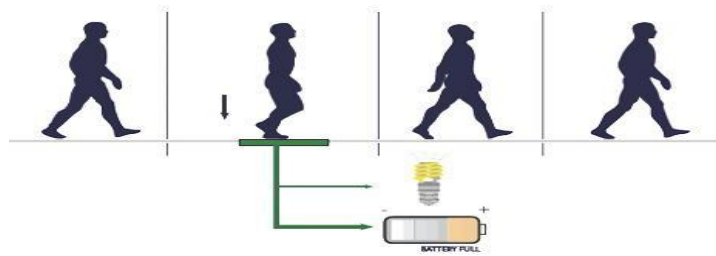


Fig 4.2: Schematic Representation

investigated. Two possible connections were tested - parallel and series connections. The parallel connection did not show significant increase in the voltage output. With series connection, additional piezo-film results in increased of voltage output but not in linear proportion. So here a combination of both parallel and series connection is employed for producing 40V voltage output with high current density. From battery provisions are provided to connect dc load. An inverter is connected to battery to provide provision to connect AC load. The voltage produced across the tile can be seen in a LCD.

Circuit Connections :

CHAPTER 5

RESULT AND DISCUSSIONS

5.1 Result

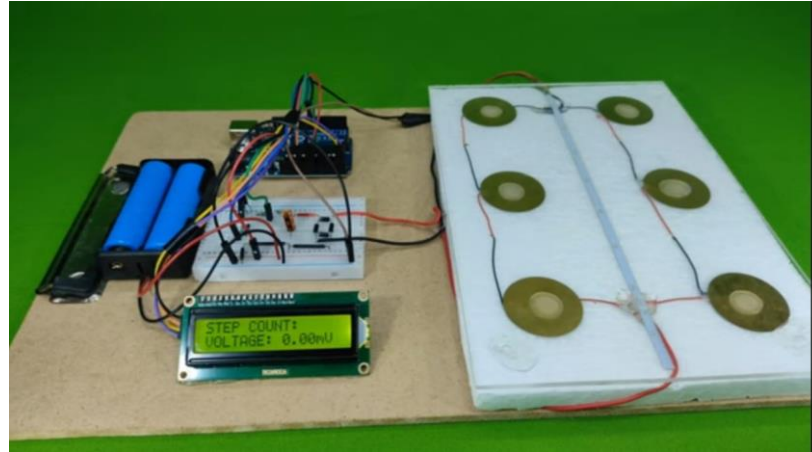


Fig 5.1 : Project Prototype

A piezo tile capable of generating 40V has been devised. Comparison between various piezo electric material shows that PZT is superior in characteristics. Also, by comparison it was found that series- parallel combination connection is more suitable. The weight applied on the tile and corresponding voltage generated is studied and they are found to have linear relation. It is especially suited for implementation in crowded areas. This can be used in street lighting without use of long power lines.

CHAPTER 6

ADVANTAGES AND DISADVANTAGES

6.1 Advantages

- Everyday wasted energy can be put to good use.
- Piezoelectric sensor is cheap and easily available.
- It is a clean source of energy which does not pollute the environment.
- No fossil fuels needed.
- Battery is used to store generated power.
- Low maintenance.
- No moving parts.
- Self-generating, no external power required.

6.2 Applications

- Can be implemented in populated metropolitan cities with a lot of footfall
- Can be installed in Railway stations, populated streets, School and College corridors.
- Power generated can be used for Street lighting.
- Footstep power generated can be used in emergency power failure situations.
- Generated power can be used for AC/DC loads.

6.3 Disadvantages

- Only applicable in high density areas.
- Power generated from piezoelectric sensor very less.
- Power generated is dependent on the pressure exerted on the sensor.

CHAPTER 7

CONCLUSION

A piezoelectric energy harvester has been simulated and designed experimentally. A constantly increasing impulse strain is applied every time to the entire unit. It has been a great experience to harvest the electrical energy from mechanical strain. The equivalent circuit model is developed in MATLAB and the expected result is obtained. The developed energy harvester can be applied to supply low powered electronics like wireless sensors, bugging devices, weather monitoring devices, aircraft power supply and many more low powered MEMS (Micro electromechanical Systems) devices. There is a wide scope of improvement of this type of harvesting technique because of increased demand of portable micro powered electronics. The all round development of self powered electronics depends upon the highly efficient energy harvesting systems. Some improvements have been done in this model to reduce the voltage drop at rectifier stage using dedicated IC. Further improvements may be done to minimize the loss and to accumulate the optimum power.

CHAPTER 8

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